NAO: 3306

16-09-2025

IN-MEXICO PROGRAM BACKEND DEVELOPER CERTIFICATION

PLAN C2 S2

Back End in Java for Information Processing

Valeria Anel Duque Salgado

Contents:

No table of contents entries found.

User Story	Requirements
As a developer (Beatriz): **I want to understand the current manual JSON integration process** **So that I can design an automation solution that meets the department's needs**	Meet with Hortensia to document the manual process. Research Java libraries for handling JSON and CSV. Create an initial technical specification document.
As a developer: **I want to set up the development environment and GitHub repository** **So that I can maintain version control and collaborate effectively**	Create a GitHub repository for the project. Configure the Java environment with required libraries. Define branching strategy (main, develop, feature).
As a developer: **I want to implement and test modules that parse JSON into CSV** **So that I can ensure data is correctly transformed and reusable**	Create a function to read and parse JSON files. Create a function to write data in CSV format. Perform unit tests to validate the conversion.
As a developer: **I want to integrate multiple JSON files into a single CSV output** **So that the department has consolidated scientific production data**	Merge multiple JSON files into a single CSV. Validate data consistency and accuracy. Perform integration tests using real cases.

As a developer **I want to document the code, process, and usage guidelines** **So that the Scientometrics Department can maintain and extend the solution**	Write technical documentation (algorithms, libraries, design). Prepare a user guide for Érika and her team. Deliver the final version in GitHub with a complete README.

Requirements	Stages	Time estimation	Deliverables
Meet with Hortensia to document the manual process. **Research Java libraries for handling JSON and CSV.** **Create an initial technical specification document.**	1. Schedule and conduct meeting with Hortensia. 2. Take notes and map the manual integration workflow. 3. Research suitable Java libraries (e.g., Jackson, Gson, OpenCSV). 4. Draft technical documentation with findings.for branch B.	1–2 days	Process documentation. Technical specification document.
Create a GitHub repository for the project. **Configure the Java environment with required libraries.** **Define branching strategy (main, develop, feature).**	1. Install/configure Java IDE and libraries. 2. Initialize GitHub repository and push initial commit. 3. Define and document branching workflow.	0.5–1 day	GitHub repository with README initialized. Configured development environment. Documented Git branching strategy.
Create a function to read and parse JSON files. **Create a function to write data in CSV format.**	1. Implement JSON reader module. 2. Implement CSV writer module. 3. Write unit tests for each module. 4. Run tests and debug	2–3 days	JSON parser function. CSV writer function. Unit test results.

Perform unit tests to validate the conversion.	issues.		
Merge multiple JSON files into a single CSV. **Validate data consistency and accuracy.** **Perform integration tests using real cases.**	1. Develop integration logic for multiple JSON inputs. 2. Test with sample datasets. 3. Validate merged CSV output for accuracy. 4. Perform end-to-end testing with real department data.	2–3 days.	Integration algorithm. Consolidated CSV output. End-to-end test report.
Write technical documentation (algorithms, libraries, design). **Prepare a user guide for Érika and her team.** **Deliver the final version in GitHub with a complete README.**	1. Write code documentation (comments, Javadoc). 2. Prepare user guide (steps to use the program). 3. Update GitHub repository with README and instructions. 4. Deliver final package to the department.	1–2 days	Technical documentation. User guide. Final GitHub repository with README.